

Zeke Mohammed

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Engineering Physics student with hands-on experience in embedded systems, FPGA development, and computational research. Completing accelerated 3-year degree while managing concurrent research and teaching roles.

EDUCATION

Fordham University

Bronx, NY

Bachelor of Science in Engineering Physics, Electrical Concentration; Minor in Mathematics; GPA: 3.31 Expected May 2026

- Completing degree in 3 years while working 15–20 hours/week in research and teaching roles
- Sarah Colman Klemmer Research Fellow, Sigma Xi Honor Society, CSTEP Scholar
- Coursework: Electricity & Magnetism I/II, Quantum Mechanics, Classical Mechanics, Circuits, Experimental Techniques in Engineering, Linear Algebra, Multivariable Calculus

TECHNICAL SKILLS

Embedded Systems & FPGA: STM32, Raspberry Pi, Arduino, Xilinx Artix-7 FPGA, Verilog, C/C++, bare-metal programming

Software & Tools: Python, MATLAB, Java, Vivado, Git/GitHub, Linux, Flask, pandas, NumPy, matplotlib, LaTeX, SQL

Protocols & Interfaces: CAN bus, I2C, SPI, UART, PWM, OBD-II

Hardware & Test: Oscilloscope, multimeter, signal analysis, PCB assembly/rework (SMD/through-hole), soldering, electrical schematics

CAD & Fabrication: Fusion 360, SolidWorks, Blender, 3D printing

TECHNICAL PROJECTS

FPGA Real-Time Image Processing Accelerator | Verilog, Vivado, Python

June 2025 – Present

- Designed pipelined convolution engine on Xilinx Artix-7 with systolic array architecture and fixed-point arithmetic, achieving 50x speedup over CPU baseline; built Python-UART verification interface and VGA output for real-time debugging

Flight Controller with Power Distribution System | STM32, C, PID Control

November 2024 – May 2025

- Developed bare-metal STM32F4 firmware with 1kHz control loop, cascaded PID controllers, and sensor fusion via I2C/SPI; designed power distribution with 11.1V LiPo, ESC/PWM motor control, and overcurrent protection; achieved stable hover with <200ms response

Multi-Sensor Data Acquisition System (M3) | Python, Raspberry Pi, Flask

October 2023 – April 2024

- Integrated 12+ sensors across CAN bus, I2C, and UART with 50Hz real-time sampling, automated logging, and Flask telemetry dashboard; validated <2% measurement error across 50+ hours of field testing

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern

June 2024 – August 2024

Hyzon Motors

Rochester, NY

- Characterized 200kW hydrogen fuel cell stack by executing V-I polarization curve tests across varied operating conditions; post-processed data in Python to isolate ohmic, activation, and mass transport loss contributions
- Supported high-voltage system validation (400–800V DC bus): performed insulation resistance, hipot, and continuity testing on HV harnesses; verified isolation monitoring device (IMD) responses and documented results against design specifications
- Assisted in monitoring battery management system (BMS) telemetry over CAN bus during fuel cell-hybrid integration testing; logged and analyzed state-of-charge and thermal data to support system validation

Undergraduate Researcher, Lattice QCD

August 2024 – Present

Fordham University Physics Department

Bronx, NY

- Built Python simulation framework for large-scale sparse matrix operations (6144×6144) using BiCGSTAB/GMRES solvers; generalized from $SU(2)$ to $SU(N)$ gauge groups with 3x runtime improvement; validated across 200+ Monte Carlo configurations

Teaching Assistant, Lab Instructor, and Grader

June 2025 – Present

Fordham University Physics Department

Bronx, NY

- Instruct 30+ students in experimental physics; evaluate 120+ lab reports monthly with technical feedback on methodology and error analysis

Lab Manager, Instructional Technology

August 2023 – Present

Fordham University Educational Technology

Bronx, NY

- Manage IT infrastructure for 50-workstation teaching lab; maintain 99%+ uptime and reduced deployment time by 40% through improved procedures

HVAC Electrical Technician

June 2023 – September 2023

Airflow Cooling & Heating Inc.

Queens, NY

- Troubleshoot electrical faults in HVAC systems; installed thermostat control systems and performed wiring in compliance with electrical codes

Freelance Electronics Technician

January 2020 – Present

Self-Employed

Queens, NY

- Served 50+ clients: console modifications (precision soldering, firmware flashing), custom PC builds, and component-level hardware repair and troubleshooting

LEADERSHIP AND ACTIVITIES

Fordham Rocket Propulsion Lab, Co-Founder (Sept 2024 – Present): Revived inactive rocketry club; design propulsion systems, perform thrust calculations, trajectory modeling, and safety analysis; CAD in Fusion 360 and 3D printing fabrication

Physics and Math Tutor, Knack (June 2024 – Present): 5.0/5.0 rating across 50+ sessions | **Fordham Physics Society** (Sept 2023 – Present): Hands-on projects including Arduino robotics and electromechanical systems